

INTERNSHIP REPORT
ON
AI/ML TRAINING AND INTERNSHIP
TRL FUTUREX

Type: Training and Internship
Period: 60 days (July 2026 – Sep 2026)

BACHELOR OF TECHNOLOGY
Computer Science and Engineering

Submitted By:
Rahul Kumar
CU ID: 233025235
4th Year / 7th Semester
Section - B

Submitted To: Mr. Vaibhav Kumar
(Internship Evaluation Faculty)



DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
COLLEGE OF SMART COMPUTING, COER UNIVERSITY
ROORKEE, UTTARAKHAND

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to **TRL Futurex** for providing me with the opportunity to undertake an internship and gain valuable practical exposure in the field of **Artificial Intelligence and Machine Learning**.

I am deeply thankful to **Ms. Saloni S.** (Trainee Mentor), for her constant guidance, support, encouragement, and valuable feedback throughout the internship. Her suggestions and knowledge helped me improve my technical understanding and successfully complete this project.

I would also like to express my sincere gratitude to **COER University** and the **Department of Computer Science & Engineering** for providing me with the opportunity to undertake this internship as part of my academic and professional development. I am equally grateful to **Mr. Vaibhav Kumar** for his academic guidance and support during the preparation of this internship report.

Finally, I would like to thank everyone who supported and encouraged me throughout the internship and contributed to making this experience a valuable learning opportunity.

Rahul Kumar
CU ID: 233025235
B.Tech – Computer Science & Engineering
COER University

OFFER LETTER



INTERNSHIP OFFER LETTER

Dear Rahul Kumar,

Date: 08/07/2026

We are pleased to extend to you an offer for the position of **Programmer Analyst Intern** at **TRL Futurex**. We believe that your skills and enthusiasm will make a valuable contribution to our team. This letter outlines the details of your internship with us.

Internship Details:

- **Designation:** Programmer Analyst Intern
- **Internship:** 2 Months AI / ML Training & Internship
- **Duration:** 2 Months
- **Start Date:** 08/07/2026
- **End Date:** 08/09/2026
- **Location:** Virtual/Physical Internship
- **Work Hours:** Monday to Friday, 9:30 AM - 6:30 PM



Roles and Responsibilities:

As a Programmer Analyst Intern, you will be expected to:

- Assist in the design, development, and testing of software applications.
- Collaborate with senior developers and project managers to understand and implement project requirements.
- Analyze system issues and provide potential solutions.
- Write and maintain clean, efficient, and well-documented code.
- Participate in code reviews and team meetings to discuss progress and challenges.
- Support various projects as required during the internship period.

We are excited about the potential contributions you will make during your internship with us.



Authorized Signatory

Address: 406, Sai Sapphire Square, Chandkheda, Ahmedabad, Gujarat 382424 **Email:** info@trifuturex.com

ATTENDANCE SHEET

Name: Rahul Kumar			
Company: TRL FUTUREX			
Role: Trainee (Intern)			
DATE	P/A	DATE	P/A
08-07-2026	P	07-08-2026	P
09-07-2026	P	08-08-2026	P
10-07-2026	P	10-08-2026	P
11-07-2026	P	11-08-2026	P
13-07-2026	P	12-08-2026	P
14-07-2026	P	13-08-2026	P
15-07-2026	P	14-08-2026	P
16-07-2026	P	17-08-2026	P
17-07-2026	P	18-08-2026	P
18-07-2026	P	19-08-2026	P
20-07-2026	P	20-08-2026	P
21-07-2026	P	21-08-2026	P
22-07-2026	P	22-08-2026	P
23-07-2026	P	24-08-2026	P
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28-07-2026	P	29-08-2026	P
29-07-2026	P	31-08-2026	P
30-07-2026	P	01-09-2026	P
31-07-2026	P	02-09-2026	P
01-08-2026	P	03-09-2026	P
03-08-2026	P	05-09-2026	P
04-08-2026	P	07-09-2026	P
05-08-2026	P	08-09-2026	P
06-08-2026	P		

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INTRODUCTION

Artificial Intelligence (AI) and Machine Learning (ML) have become important areas of modern computer science, with applications in education, healthcare, finance, business, communication, and many other fields. Machine Learning enables computer systems to learn patterns from data and use those patterns to make predictions or support decision-making. Developing practical knowledge in these areas requires not only understanding theoretical concepts but also applying them through programming, data analysis, and machine-learning projects.

As part of my academic and professional development, I undertook an **AI / ML Training & Internship at TRL Futurex**. The internship provided practical exposure to Python programming, data handling, statistical analysis, data visualization, data preprocessing, supervised learning, and unsupervised learning. The training was supported by practical assignments and projects, allowing me to apply the concepts covered during the learning sessions. During the internship, Python was used as the primary programming language for implementing different tasks. Libraries and tools such as **NumPy, Pandas, Matplotlib, Seaborn, SciPy, and Scikit-learn** were used for numerical computation, data manipulation, statistical analysis, visualization, and machine-learning model development. The practical work gradually progressed from basic Python programming and student-data processing to more advanced machine-learning tasks.

The internship also provided experience in working with structured datasets and following a systematic data-analysis process. This included identifying and handling missing values, removing duplicate records, separating features and target variables, performing statistical analysis, visualizing relationships between variables, and preparing data for machine-learning models.

As part of the machine-learning activities, different approaches were explored for prediction and classification problems. Regression and classification techniques were applied to student-related data, while clustering techniques were used to identify groups of students with similar characteristics. These activities helped develop a practical understanding of how machine-learning methods can be selected and applied according to the nature of a problem.

One of the important practical areas of the internship was the development of a **Student Performance Prediction System**, where machine-learning techniques were applied to student-related attributes to predict performance. The internship also included work related to feature engineering and student clustering, providing exposure to both supervised and unsupervised learning approaches.

Overall, the internship served as a bridge between academic learning and practical implementation. It helped me understand the workflow involved in an AI/ML task, from writing Python programs and preparing data to developing models and interpreting their results.

OBJECTIVES OF THE INTERNSHIP

The primary objective of the internship was to gain practical knowledge and hands-on experience in the field of **Artificial Intelligence and Machine Learning** and to understand how programming, data analysis, and machine-learning techniques can be applied to practical problems. The internship was designed to strengthen my Python programming skills and develop the ability to write structured programs using fundamental programming concepts such as variables, data structures, functions, and data manipulation techniques.

Another important objective was to develop a strong understanding of the fundamentals of **Artificial Intelligence and Machine Learning** and to learn how data-driven techniques can be used for prediction, classification, and analysis. The internship provided practical exposure to libraries such as **NumPy and Pandas**, which helped me understand numerical computation, data manipulation, and working with structured datasets. I also aimed to develop skills in data analysis and visualization using **Matplotlib, Seaborn, and SciPy** to study patterns, distributions, relationships, and statistical characteristics within datasets.

The internship also focused on understanding the complete **data preprocessing workflow**, including identifying and handling missing values, removing duplicate records, separating features and target variables, and preparing datasets for machine-learning applications. Through practical assignments, I gained exposure to **supervised learning techniques**, particularly regression and classification, and learned how these techniques can be applied to prediction problems. I also developed an understanding of **unsupervised learning**, particularly K-Means clustering, for grouping data points based on their characteristics.

A further objective was to develop practical knowledge of **feature engineering and model evaluation**. Feature engineering helped me understand how useful information can be derived from existing data, while model evaluation introduced me to methods for assessing the performance of machine-learning models. These concepts were applied through practical student-data projects and assignments.

The internship also aimed to improve my **analytical thinking, problem-solving, and independent working abilities**. By completing assignments and projects individually, submitting the work through the prescribed platform, and incorporating feedback received during the training, I gained experience in following a structured approach to practical tasks. The overall objective was to bridge the gap between academic knowledge and practical implementation while developing a strong foundation for further learning and professional work in Artificial Intelligence and Machine Learning.

ORGANIZATION PROFILE: TRL FUTUREX

Company Name: TRL Futurex

Industry: Information Technology

Address: 406, Sai Sapphire Square, Chandkheda, Ahmedabad, Gujarat – 382424, India

Specialization

TRL Futurex provides training and practical learning opportunities in **Artificial Intelligence and Machine Learning**. The internship program focuses on developing practical technical skills through live training sessions, assignments, and project-based activities. During the internship, the training covered Python programming, data handling, statistical analysis, data visualization, data preprocessing, supervised and unsupervised machine learning, feature engineering, and model evaluation.

Background

TRL Futurex provided a structured technical learning environment through its **AI / ML Training & Internship** program. The program was designed to provide practical exposure to programming and machine-learning concepts through hands-on assignments and projects. As part of the internship, participants progressed from Python fundamentals and data processing to the implementation of machine-learning techniques for practical problems.

The official internship offer letter confirms that the internship was offered for the position of **Programmer Analyst Intern** under the **2 Months AI / ML Training & Internship** program. It also describes responsibilities related to software development, testing, analysing system issues, developing solutions, maintaining code, and supporting projects.

Mission / Goal

The primary goal of the internship program was to provide practical exposure to **Artificial Intelligence and Machine Learning** and help participants develop the technical and analytical skills required to work with data-driven applications. The program emphasized learning through practical implementation, assignments, projects, evaluation, and feedback.

Through this structured approach, the internship aimed to bridge the gap between theoretical knowledge and practical application by enabling learners to apply programming, data analysis, and machine-learning concepts to real-world-oriented problems.

INTERNSHIP TRAINING AND WORK METHODOLOGY

The internship at **TRL Futurex** was conducted in an online mode and followed a practical and structured learning approach. The training consisted of **live training sessions, individual assignments, and project-based activities**. This methodology provided an opportunity to understand the concepts introduced during the training and apply them through practical programming and machine-learning tasks.

The training began with the fundamentals of **Artificial Intelligence, Machine Learning, and Python programming**. Python was used as the primary programming language for the practical activities. The initial assignments focused on developing programming fundamentals and understanding the use of Python data structures and functions. These concepts were applied through practical tasks involving student information and student-related data.

The training then progressed towards **numerical computing and data handling** using libraries such as **NumPy and Pandas**. Practical assignments involved creating, organizing, and processing student data. Numerical operations and statistical calculations were performed to understand how Python libraries can be used efficiently for handling datasets.

The next stage focused on **data analysis and visualization**. Statistical techniques were used to examine student-related data, including measures such as mean, median, mode, variance, standard deviation, range, and interquartile range. Correlation analysis and outlier detection were also performed. Data visualization techniques using **Matplotlib and Seaborn** were used to represent patterns and relationships within the dataset through different types of graphs and plots.

The training further covered **data preprocessing**, which is an important stage before applying machine-learning algorithms. Practical work included identifying missing values, handling incomplete data, detecting and removing duplicate records, separating features and target variables, and preparing datasets for machine-learning models. Feature engineering was also introduced to create meaningful information from existing data.

After completing the data preparation activities, the training progressed to **machine-learning techniques**. Supervised learning methods were applied to student-performance-related problems. **Linear Regression** was used for numerical prediction, while **Logistic Regression** was used for classification. The training also introduced unsupervised learning through **K-Means Clustering**, where student data was grouped according to selected performance-related characteristics.

The assignments were completed **individually** and submitted through the prescribed platform. The submitted work was reviewed and feedback was provided, which helped in identifying areas for improvement and strengthening the understanding of the concepts. This process encouraged independent learning and practical problem-solving.

INTRODUCTION TO ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

Artificial Intelligence (AI) is one of the major areas of modern computer science that focuses on developing systems capable of performing tasks that generally require human intelligence. These tasks may include learning from information, recognizing patterns, making predictions, understanding data, and supporting decision-making. AI technologies are increasingly being applied in areas such as education, healthcare, finance, transportation, communication, and business.

Machine Learning (ML) is an important area of Artificial Intelligence that enables computer systems to learn patterns from data and use those patterns to make predictions or decisions. Instead of developing a separate rule for every possible situation, machine-learning algorithms learn from available data and develop a model that can be used on new or unseen data.

During the internship at **TRL Futurex**, I developed a practical understanding of the basic concepts of Artificial Intelligence and Machine Learning. The training introduced the role of data in machine-learning applications and demonstrated how programming and data analysis techniques can be combined to develop predictive solutions.

Types of Machine Learning

The internship training introduced the concepts of **supervised learning and unsupervised learning**. In supervised learning, a model is trained using data containing known input features and corresponding output values. This approach can be used for both regression and classification problems. In the practical assignments, supervised learning techniques were applied to student-related data for predicting performance and classifying students.

In unsupervised learning, the model works with data without predefined output labels and attempts to identify patterns or groups within the data. **K-Means Clustering** was used during the practical work to group students based on selected performance-related characteristics.

Role of Data in Machine Learning

Data is a fundamental component of any machine-learning application. The quality, structure, and preparation of data have an important effect on the performance of a model. Therefore, the internship placed significant emphasis on understanding data before applying machine-learning algorithms.

The practical activities included data processing, statistical analysis, visualization, identification of missing values, removal of duplicate records, separation of features and target variables, and

preparation of datasets for machine-learning models. These activities demonstrated that successful machine-learning implementation involves more than simply selecting an algorithm.

Machine Learning Workflow

The practical learning followed a systematic workflow:

Data → Data Analysis → Data Preprocessing → Feature Engineering → Model Training → Prediction/Clustering → Model Evaluation

This workflow was gradually applied throughout the internship. The practical work started with basic Python programming and student-data processing and progressed towards statistical analysis, data preprocessing, supervised learning, and unsupervised learning.

Overall, the internship provided a practical foundation in Artificial Intelligence and Machine Learning and helped me understand how data-driven techniques can be implemented using Python and related libraries. The concepts learned in this stage formed the foundation for the subsequent practical work involving **NumPy, Pandas, Matplotlib, Seaborn, SciPy, and Scikit-learn**.

PYTHON PROGRAMMING AND MATHEMATICAL FOUNDATIONS

Python was used as the primary programming language during the internship for implementing practical tasks related to data processing and machine learning. The training began with Python fundamentals, which provided the programming foundation required for working with datasets, performing calculations, analysing information, and developing machine-learning models.

The Python programming activities included fundamental concepts such as **variables, data types, operators, conditional statements, loops, lists, dictionaries, functions, and input/output operations**. These concepts were applied through practical assignments. One of the initial activities involved developing a **Student Information System**, in which student details such as name, age, and marks were stored and displayed using Python data structures. Functions were also used to perform specific operations, such as calculating the average marks of students.

Mathematical Foundations

Mathematics and statistics play an important role in data analysis and machine learning because they provide the methods required to understand datasets, measure relationships between variables, and evaluate the behaviour of models. During the internship, several basic statistical concepts were applied to student-related data.

The **mean** represents the average value of a dataset and can be calculated as:

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$$

where x_i represents an individual observation and n represents the total number of observations.

The **median** is the middle value when observations are arranged in ascending or descending order, while the **mode** represents the value that occurs most frequently. These measures help describe the central tendency of a dataset.

The **variance** measures how far individual observations are spread from the mean. For a population, it can be expressed as:

$$\sigma^2 = \frac{\sum_{i=1}^n (x_i - \mu)^2}{n}$$

where μ is the population mean. The **standard deviation** is the square root of variance:

$$\sigma = \sqrt{\sigma^2}$$

A smaller standard deviation indicates that observations are generally closer to the mean, whereas a larger value indicates greater variation in the data.

The **range** provides a simple measure of dispersion and is calculated as:

$$\text{Range} = \text{Maximum Value} - \text{Minimum Value}$$

Another useful measure covered during the statistical analysis was the **Interquartile Range (IQR)**. It represents the difference between the third quartile and first quartile:

$$IQR = Q_3 - Q_1$$

The IQR can also be used to identify potential outliers. Observations below $Q_1 - 1.5(IQR)$ or above $Q_3 + 1.5(IQR)$ may be considered potential outliers.

Correlation

Correlation is used to measure the degree and direction of association between variables. The **Pearson correlation coefficient** measures the strength of a linear relationship and can be represented as:

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}$$

The value of r generally lies between **-1 and +1**. A positive value indicates a positive relationship, a negative value indicates an inverse relationship, and a value close to zero indicates a weak linear relationship.

The internship activities also introduced **Spearman correlation**, which is based on the ranks of observations and is useful when the relationship between variables may not be strictly linear.

Statistical Analysis and Machine Learning

These mathematical concepts provided a foundation for understanding and preparing data before applying machine-learning algorithms. Statistical measures helped in describing the dataset, while correlation analysis helped examine relationships between variables. Outlier detection techniques helped identify unusual observations that could influence analysis or model performance.

NumPy for Numerical Computation

NumPy was used to perform numerical operations efficiently. It provided practical experience in working with numerical arrays and performing calculations on student-related data. The use of NumPy demonstrated how mathematical operations can be implemented programmatically instead of being calculated manually.

DATA PREPROCESSING AND FEATURE ENGINEERING

Data preprocessing is an important stage in the machine-learning workflow because the quality of the input data directly affects the reliability of analysis and model development. During the internship, practical activities were performed to understand how raw student-related data can be examined, cleaned, transformed, and prepared before applying machine-learning algorithms.

The first step in preprocessing was to **inspect the dataset** and understand the available variables and their data types. This helped identify the features that could be used for further analysis and modelling. Student-related attributes such as **Study Hours, Attendance, Previous Marks, and other performance-related information** were considered during the practical work.

Handling Missing Values

Missing values can occur when information is unavailable or has not been recorded. If missing values are not handled appropriately, they may affect statistical analysis and machine-learning models. Therefore, identifying missing values was an important part of the preprocessing process. The dataset was examined to determine whether incomplete observations were present, after which appropriate preprocessing could be applied.

Duplicate Data

Duplicate records can result in the same observation being considered more than once and may affect the analysis. The preprocessing process therefore included checking for duplicate records and removing them where necessary. This helped maintain a cleaner and more reliable dataset for subsequent analysis.

Feature and Target Separation

Before developing a machine-learning model, the available data needs to be divided into **input features and target variables**. Features represent the information provided to the model, while the target represents the value that the model is expected to predict.

For the student-performance-related activities, variables such as **Study Hours, Attendance, and Previous Marks** were used as relevant input information. Depending on the prediction task, student marks or performance categories could serve as the target variable.

Feature Engineering

Feature engineering involves creating useful information from existing variables to improve the representation of a dataset. During the internship, derived student-performance features were considered in the practical machine-learning activities.

For example, in the clustering activity, features such as **Total Marks, Average Marks, and Performance Percentage** were created from available student-performance information. These derived features provided additional ways to represent student performance and supported further analysis.

The average marks can be represented mathematically as:

$$\text{Average Marks} = \frac{\text{Total Marks}}{\text{Number of Subjects}}$$

Similarly, a percentage can be calculated using:

$$\text{Performance Percentage} = \frac{\text{Marks Obtained}}{\text{Maximum Marks}} \times 100$$

Such derived features can make the information easier to interpret and provide additional variables for machine-learning applications.

Importance of Preprocessing

Data preprocessing helps ensure that the dataset is in a suitable form for analysis and modelling. Cleaning the data, checking for missing values and duplicates, separating features from targets, and creating meaningful features can improve the quality and consistency of the information provided to a machine-learning algorithm.

The preprocessing activities performed during the internship helped me understand that machine learning does not begin directly with model training. **Understanding and preparing the data is an essential part of the overall process.** The practical experience gained through these activities provided a foundation for the subsequent implementation of regression, classification, and clustering techniques.

Overall, the data preprocessing and feature engineering stage improved my ability to work systematically with datasets and understand how raw information can be transformed into a useful form for machine-learning applications.

SUPERVISED MACHINE LEARNING: REGRESSION AND CLASSIFICATION

Supervised Machine Learning is a type of machine learning in which a model learns from a dataset containing known input features and corresponding target values. The objective is to learn the relationship between the input variables and the target variable so that the trained model can make predictions for new data. During the internship, supervised learning techniques were applied to student-related data to understand how machine-learning models can be used for both numerical prediction and classification.

A supervised learning problem generally consists of **features, a target variable, training data, and testing data**. The features provide information to the model, while the target represents the expected output. The available dataset is commonly divided into training and testing portions. The training data is used to learn the relationship between the features and target, while the testing data is used to evaluate how well the trained model performs on unseen observations.

Linear Regression

Linear Regression is a supervised learning algorithm used for predicting a continuous numerical value. It assumes a relationship between the input variables and the target variable and attempts to find a suitable linear equation representing that relationship.

For a simple linear regression problem, the relationship can be represented as:

$$y = \beta_0 + \beta_1 x$$

where y represents the predicted value, x represents the input feature, β_0 represents the intercept, and β_1 represents the coefficient of the feature.

When multiple features are involved, the equation can be extended as:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n$$

In the internship project related to **student performance prediction**, Linear Regression was used for predicting a numerical performance value based on relevant student-related features. This provided practical experience in preparing the data, selecting features, training the model, and generating predictions.

Logistic Regression

Logistic Regression is a supervised learning algorithm commonly used for classification problems. Unlike Linear Regression, which predicts a continuous numerical value, Logistic Regression estimates the probability of an observation belonging to a particular class.

The logistic function can be expressed as:

$$P(y = 1) = \frac{1}{1 + e^{-z}}$$

where:

$$z = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n$$

The output probability can then be used to assign an observation to an appropriate class according to the classification threshold.

During the internship, Logistic Regression was applied to a **student performance classification problem**, where student performance could be classified into categories such as **Pass or Fail**. This practical activity helped demonstrate how classification algorithms can be used to make decisions based on multiple input features.

Model Training and Prediction

The general implementation process involved preparing the dataset, separating the features and target variable, dividing the data into training and testing sets, training the selected model, and using the trained model to generate predictions. The predictions were then evaluated using appropriate performance measures.

The practical implementation of both regression and classification helped me understand that the choice of algorithm depends on the type of problem being solved. **Linear Regression** is suitable when the objective is to predict a numerical value, whereas **Logistic Regression** can be used when the objective is to classify observations into predefined categories.

Overall, the supervised learning activities provided practical experience in applying machine-learning algorithms to student-related data and helped establish an understanding of the complete process from **data preparation to model training, prediction, and evaluation**.

UNSUPERVISED LEARNING AND K-MEANS CLUSTERING

Unsupervised Machine Learning is a type of machine learning in which the model works with data without predefined target labels. Instead of predicting a known output, the objective is to identify hidden patterns, structures, or groups within the available data. During the internship, **K-Means Clustering** was introduced as an unsupervised learning technique and was applied to student-performance-related data.

K-Means Clustering

K-Means is a clustering algorithm that divides a dataset into a predefined number of groups, known as **clusters**. Each cluster is represented by a central point called a **centroid**. Data observations are assigned to the cluster whose centroid is closest according to a distance measure, commonly Euclidean distance.

The Euclidean distance between two points can be represented as:

$$d(x,c) = \sqrt{\sum_{j=1}^n (x_j - c_j)^2}$$

where x represents a data point and c represents a cluster centroid.

The basic K-Means process consists of selecting the number of clusters, initializing centroids, assigning observations to their nearest centroid, recalculating the centroids, and repeating the process until the cluster assignments become stable or the algorithm reaches its stopping condition.

Application to Student Data

As part of the internship practical work, K-Means clustering was applied to student-related data to identify groups of students with similar performance characteristics. Before clustering, relevant information was prepared through feature engineering. Features such as **Total Marks, Average Marks, and Performance Percentage** were derived from the available student-performance information.

For the clustering activity, different values of **K = 2, 3, 4, and 5** were tested. This helped in understanding how changing the number of clusters affects the grouping of students. The clustering process considered relevant performance-related features, including **Average Marks and Attendance**, to identify similarities among students.

Evaluation of Clusters

Selecting an appropriate number of clusters is an important part of K-Means clustering. During the practical activity, **inertia** and the **Silhouette Score** were used to compare different cluster configurations.

Inertia represents the within-cluster sum of squared distances between observations and their corresponding cluster centroids. A lower inertia generally indicates that observations are closer to their assigned centroids. However, increasing the number of clusters can naturally reduce inertia, so it should not be considered alone.

The **Silhouette Score** provides an indication of how well observations fit within their assigned clusters compared with other clusters. A higher score generally indicates better-defined and more separated clusters.

By comparing the results for different values of K, the clustering configurations could be analysed and interpreted based on the characteristics of the student data.

Importance of Clustering

K-Means clustering provided practical exposure to a different type of machine-learning problem from the regression and classification activities. While supervised learning uses known target values, clustering attempts to discover groups within the data without predefined labels.

The activity helped demonstrate how unsupervised learning can be used to explore student-performance patterns and identify groups of students with similar characteristics. It also provided practical experience in **feature engineering, clustering, evaluation using inertia and Silhouette Score, and interpretation of machine-learning results**.

Overall, the K-Means activity strengthened my understanding of unsupervised learning and demonstrated how clustering techniques can be applied to analyse structured student-related data.

PRACTICAL ASSIGNMENTS AND MODEL EVALUATION

The practical assignments completed during the internship were designed to provide a gradual understanding of **Python programming, data processing, statistical analysis, data preprocessing, and machine learning**. The assignments progressed from basic programming tasks to more advanced applications involving prediction and clustering. This progression helped in developing practical skills by applying the concepts learned during the training sessions.

The initial practical work focused on **Python programming and student data management**. A Student Information System was developed to store and display basic student information such as name, age, and marks. This activity provided practice with Python variables, data structures, input/output operations, and functions. The use of functions also helped in organizing the program and performing specific operations such as calculating average marks.

The next stage involved **student data processing using Python and NumPy**. Numerical student information was processed to perform calculations and generate useful summaries. This activity provided practical experience in using NumPy for numerical operations and strengthened the understanding of working with structured student-related data.

The internship then progressed towards **statistical analysis and data visualization**. Statistical measures such as mean, median, mode, variance, standard deviation, range, and interquartile range were calculated to understand the characteristics of the dataset. Correlation analysis was also performed to examine relationships between variables. Visualization techniques using Matplotlib and Seaborn were used to represent distributions, relationships, and patterns within the data.

Another practical assignment focused on **complete data preprocessing**. The work involved examining the dataset, checking for missing values, identifying duplicate records, separating features and target variables, and preparing the data for machine-learning applications. Feature engineering was also performed to create useful derived features from existing student-performance information.

The later assignments introduced **machine-learning techniques**. Linear Regression was used for numerical prediction, while Logistic Regression was applied to a classification problem involving student performance. These activities provided practical experience in preparing data, dividing it into training and testing sets, training models, generating predictions, and evaluating their performance.

The internship also included an **unsupervised learning activity using K-Means Clustering**. Different values of K were tested to understand how students could be grouped according to selected performance-related characteristics. Inertia and Silhouette Score were considered for comparing different clustering configurations.

Model Evaluation

Model evaluation is an essential part of machine-learning development because it provides a way to determine how effectively a model performs on data. During the internship, different evaluation measures were considered according to the type of machine-learning task.

For regression problems, **Mean Squared Error (MSE)** and the **R² Score** were used as relevant evaluation measures. MSE measures the average squared difference between actual and predicted values and can be expressed as:

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y})^2$$

where y_i represents the actual value and $\hat{y}$ represents the predicted value.

The **R² Score** provides an indication of how well the model explains the variation in the target variable. In general, a higher R² value indicates that the model explains a greater proportion of the observed variation.

For classification, **accuracy** was considered as an evaluation measure. It represents the proportion of correctly classified observations:

$$Accuracy = \frac{\text{Correct Predictions}}{\text{Total Predictions}}$$

For the K-Means clustering activity, **Inertia and Silhouette Score** were used to compare different cluster configurations. These measures helped in understanding the compactness and separation of the resulting clusters.

Overall, the practical assignments provided a progressive learning experience, beginning with programming fundamentals and moving towards data analysis and machine-learning applications. The evaluation activities further helped me understand that developing a model is only one part of the machine-learning process; **measuring and interpreting its performance is equally important.**

STUDENT PERFORMANCE PREDICTION SYSTEM

The **Student Performance Prediction System** was one of the major practical projects undertaken during the AI/ML internship at TRL Futurex. The project was designed to apply the concepts of data preprocessing, feature selection, supervised machine learning, prediction, and model evaluation to a student-related dataset. It provided an opportunity to combine the programming and data-analysis concepts learned during the earlier stages of the internship into a practical machine-learning application.

Project Objective

The main objective of the system was to use available student-related information to develop machine-learning models capable of predicting student performance. The project involved both a **regression problem**, where a numerical performance value could be predicted, and a **classification problem**, where students could be classified according to their performance category.

The project helped demonstrate how machine-learning techniques can be applied to educational data to identify relationships between student characteristics and academic performance. It also provided practical experience in following a complete machine-learning workflow from data preparation to prediction and evaluation.

Dataset and Features

The project used student-related data containing relevant academic and behavioural attributes. Important features considered during the practical work included **Study Hours, Attendance, and Previous Marks**. These variables provide information that can potentially contribute to understanding and predicting student performance.

Before applying machine-learning algorithms, the dataset was examined and prepared for modelling. The input features were separated from the target variable so that the models could learn the relationship between the available student information and the expected output.

Data Preparation

Data preparation was an important stage of the project. The dataset was checked for issues such as missing values and duplicate records. The relevant features were then selected and organized for use in the machine-learning models.

The prepared dataset was divided into **training and testing data**. The training portion was used to train the machine-learning models, while the testing portion was kept for evaluating their

performance on previously unseen observations. This separation helps provide a more realistic assessment of how a model may perform when given new data.

Linear Regression for Performance Prediction

Linear Regression was applied as a regression technique to predict a numerical student-performance value. The model attempts to learn the relationship between the selected input features and the target value.

The general multiple linear regression equation can be represented as:

$$\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n$$

where $\hat{y}$ represents the predicted value, $x_1, x_2, \dots, x_n$ represent the input features, and the coefficients represent the contribution of the respective features to the prediction.

In the project, student-related factors such as **Study Hours, Attendance, and Previous Marks** were used as inputs for the prediction task.

Logistic Regression for Classification

The project also involved **Logistic Regression** for classification. Instead of predicting a continuous numerical value, the classification model was used to determine the student's performance category, such as **Pass or Fail**.

Logistic Regression calculates a probability for the possible class and uses the resulting probability to assign the observation to an appropriate category. This provided practical experience in understanding the difference between regression and classification problems.

Overall Project Workflow

The Student Performance Prediction System followed the following general workflow:

Student Data → Data Preprocessing → Feature Selection → Train-Test Split → Model Training → Prediction → Model Evaluation

Through this project, I gained practical experience in applying supervised machine-learning techniques to a real-world-oriented problem. The project also helped strengthen my understanding of how **Python, Pandas, NumPy, and Scikit-learn** can be used together for developing a machine-learning solution.

The project served as an important practical application of the concepts covered during the internship and provided a foundation for analysing model outputs and evaluating the effectiveness of the developed prediction system.

PROJECT IMPLEMENTATION, RESULTS AND ANALYSIS

The implementation of the **Student Performance Prediction System** involved applying the concepts learned during the internship to a structured student-related dataset. The project followed a systematic machine-learning workflow, beginning with data preparation and continuing through model development, prediction, and evaluation.

Implementation Process

The first stage of implementation involved preparing the dataset for analysis and modelling. The relevant student attributes were identified and separated into input features and target variables. Features such as **Study Hours, Attendance, and Previous Marks** were considered for predicting student performance.

The dataset was then divided into **training and testing sets**. The training data was used to allow the machine-learning models to learn relationships between the input features and the target variable. The testing data was subsequently used to evaluate the models using data that was not used during training.

The project implemented two supervised learning approaches. **Linear Regression** was used for the numerical prediction task, while **Logistic Regression** was used for the classification task. Both models were implemented using Python and the relevant machine-learning libraries.

Linear Regression Analysis

The Linear Regression model was trained using the selected student-related features. After training, the model generated predicted performance values for the testing data. The predictions were compared with the actual values to determine the effectiveness of the model.

The performance of the regression model was assessed using **Mean Squared Error (MSE)** and **R² Score**. MSE measures the average squared difference between actual and predicted values, while the R² Score indicates how effectively the model explains the variation in the target variable.

The actual numerical values obtained from the model execution should be reported directly from the project output:

Mean Squared Error (MSE): [Insert actual value]

R² Score: [Insert actual value]

These values provide a quantitative basis for analysing the performance of the Linear Regression model.

Logistic Regression Analysis

Logistic Regression was implemented for the classification component of the project. The objective was to classify students into performance categories such as **Pass or Fail** based on the selected input features.

After training the model, predictions were generated for the testing dataset. The classification results were evaluated using **accuracy**, which represents the proportion of correctly classified observations out of the total number of observations.

Classification Accuracy: [Insert actual value]

The accuracy value obtained from the actual project execution should be used in the final report rather than an estimated or assumed value.

Analysis of Results

The implementation demonstrated the practical difference between regression and classification tasks. Linear Regression produced a numerical prediction, making it appropriate for predicting a continuous performance value. Logistic Regression, on the other hand, produced a classification outcome and was therefore suitable for distinguishing between performance categories.

The project also demonstrated the importance of selecting appropriate features and preparing the dataset before model training. Factors such as **study hours, attendance, and previous academic performance** can provide useful information for analysing student performance.

The results obtained from the actual model execution should be interpreted together with the selected evaluation metrics. Rather than considering a single metric in isolation, the model should be assessed based on how well it performs on unseen testing data and whether its predictions are meaningful for the intended application.

Project Significance

The Student Performance Prediction System provided practical experience in developing a complete supervised machine-learning solution. It demonstrated the connection between **data preprocessing, feature selection, model training, prediction, and evaluation**.

The project also strengthened my understanding of using Python-based machine-learning tools to solve a practical problem. Most importantly, it provided hands-on experience in transforming a structured dataset into a machine-learning model capable of producing predictions.

Note: We should insert the **actual MSE, R^2 , and accuracy values from your project output** before finalizing this page. I don't want to fabricate results for your college report.

CHALLENGES FACED AND LEARNING OUTCOMES

Challenges Faced

During the internship, I faced several challenges while progressing from basic Python programming to practical Artificial Intelligence and Machine Learning applications. One of the initial challenges was understanding how programming concepts could be applied to practical data-related problems. Working with student-related datasets required careful attention to data organization, numerical operations, and the appropriate use of Python libraries.

Another challenge was understanding the different stages of **data preprocessing**. Identifying missing values, checking duplicate records, selecting relevant features, and separating input features from target variables required a systematic approach. I also had to understand how statistical measures and visualizations could be used to examine the characteristics and relationships present in the data.

Understanding the differences between **regression, classification, and clustering** was another important challenge. Applying Linear Regression and Logistic Regression to different types of problems required an understanding of the nature of the target variable. Similarly, working with K-Means Clustering required understanding how data points are grouped and how the appropriate number of clusters can be analysed using evaluation measures.

Interpreting **model evaluation metrics** was also an important learning challenge. Understanding measures such as MSE, R^2 Score, Accuracy, Inertia, and Silhouette Score required connecting the mathematical concepts with the actual results produced by the models.

These challenges were addressed through regular practice, reviewing the concepts introduced during the training, completing practical assignments, analysing program outputs, and applying the feedback received during the internship. This gradual process helped improve my understanding and confidence in working with AI/ML tasks.

Learning Outcomes

The internship significantly improved my practical understanding of **Python programming and Artificial Intelligence and Machine Learning**. I gained hands-on experience with Python libraries including **NumPy, Pandas, Matplotlib, Seaborn, SciPy, and Scikit-learn** and understood their applications in numerical computation, data manipulation, statistical analysis, visualization, and machine learning.

I developed an understanding of the complete **data science and machine-learning workflow**, beginning with data handling and analysis and progressing through preprocessing, feature

engineering, model training, prediction, and evaluation. Practical implementation of **Linear Regression, Logistic Regression, and K-Means Clustering** helped me understand the applications of both supervised and unsupervised learning.

The internship also strengthened my ability to perform statistical analysis using measures such as mean, median, mode, variance, standard deviation, correlation, and IQR. Working with visualizations helped me understand how graphical representations can be used to identify patterns and relationships within datasets.

The practical assignments and project work further improved my **problem-solving, analytical thinking, debugging, and independent learning skills**. Completing tasks individually and incorporating feedback helped me develop a more systematic approach to solving technical problems.

Overall, the internship helped me connect theoretical knowledge gained through academic study with practical implementation. It provided a strong foundation for further learning and development in the fields of **Artificial Intelligence, Machine Learning, Data Analysis, and Python-based application development**.

INTERNSHIP COMPLETION CERTIFICATE



Certificate of Completion

Recognizing dedication, commitment, and successful completion

This is proudly presented to

RAHUL KUMAR

for successfully completing

AI / ML Training & Internship

Completed: 08/09/2026 | Duration: 2 Months



Authorized Signatory

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www.trlfuturex.com

Reg. Office Address: 406-407, 4th Floor, Sai Sapphire Square, Near KB Royal Serenity, Chandkheda, Ahmedabad, Gujarat
382424 | **Email:** info@trlfuturex.com

CONCLUSION

The **AI / ML Training & Internship at TRL Futurex** was a valuable practical learning experience that provided me with an opportunity to apply academic concepts in a structured and practical environment. The internship helped me develop a better understanding of Artificial Intelligence and Machine Learning and the different stages involved in solving data-driven problems.

Throughout the internship, I developed practical programming skills using **Python** and gained experience with important libraries such as **NumPy, Pandas, Matplotlib, Seaborn, SciPy, and Scikit-learn**. The practical assignments helped me understand how these tools can be used for numerical computation, data processing, statistical analysis, visualization, preprocessing, and machine-learning applications.

The internship also provided hands-on experience with **supervised and unsupervised learning techniques**. The implementation of Linear Regression and Logistic Regression provided an understanding of numerical prediction and classification problems, while K-Means Clustering introduced the process of identifying groups within data without predefined labels.

The **Student Performance Prediction System** was an important practical application of the concepts learned during the internship. It provided experience in preparing student-related data, selecting relevant features, training machine-learning models, generating predictions, and evaluating model performance. Working on this project helped demonstrate how machine-learning techniques can be applied to practical problems.

The internship also helped improve my **analytical thinking, problem-solving ability, programming skills, and independent learning**. The combination of live training, practical assignments, project work, submission, and feedback provided an effective learning experience and helped strengthen my confidence in applying technical concepts.

In conclusion, the internship successfully helped bridge the gap between **theoretical academic knowledge and practical implementation**. The knowledge and experience gained during this period have provided me with a stronger foundation in Artificial Intelligence and Machine Learning and will be useful for my future academic projects, professional development, and further exploration of AI/ML technologies.